1. IMO 2025 Model Solutions

Problem 1

A line in the plane is called *sunny* if it is not parallel to any of the x-axis, the y-axis, or the line $x + y = 0$.
Let $n \geq 3$ be a given integer. Determine all nonnegative integers k such that there exist n distinct lines in the plane satisfying both of the following:

- for all positive integers a and b with $a + b \leq n + 1$, the point (a, b) lies on at least one of the lines; and
- exactly k of the n lines are sunny.

Short Answer
k = 0, 1, 3 for all n

Model Proof
Notation. For an integer $n \geq 3$ define

$$S_n = \{(a, b) \in \mathbb{Z}^2 \mid a \geq 1,\ b \geq 1,\ a + b \leq n + 1\}.$$

A line in the plane is called *sunny* if its slope is neither 0 (horizontal), ∞ (vertical), nor -1 (the slope of the line $x + y = 0$).
We have to determine all non-negative integers k for which there exist n distinct lines covering all points of S_n with exactly k of them sunny.

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1. Constructions showing that $k = 0, 1, 3$ are attainable
$k = 0$.
Take the n vertical lines $x = 1, x = 2, \ldots, x = n$. Every point $(a, b) \in S_n$ has $a \leq n$, hence lies on the line $x = a$. These lines are not sunny. Clearly there are n distinct lines, so $k = 0$ is possible for every $n \geq 3$.
$k = 1$.
For $n \geq 3$ take the $n - 1$ vertical lines $x = 1, \ldots, x = n - 1$ and the line

$$L: \text{ through } (n, 1) \text{ and } (n - 2, 2).$$

The slope of L is $\frac{2-1}{(n-2)-n} = -\frac{1}{2}$; it is not $0, \infty, -1$, so L is sunny.
- Points with $a \leq n - 1$ are covered by the vertical line $x = a$.
- The only point with $a = n$ is $(n, 1)$, which lies on L.

Thus all points of S_n are covered. Hence $k = 1$ is attainable for all $n \geq 3$.
$k = 3$.
Case $n = 3$.
S_3 consists of the six points

$$(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (3, 1).$$

Take the three sunny lines

$$\begin{aligned} \ell_1 &: \ y = x, \\ \ell_2 &: \ y = -2x + 5, \\ \ell_3 &: \ y = -\tfrac{1}{2}x + \tfrac{5}{2}. \end{aligned}$$

Their slopes are $1, -2, -\frac{1}{2}$ - none is $0, \infty, -1$, so they are sunny. One checks directly that each of the six points belongs to at least one of them. Therefore $k = 3$ is possible for $n = 3$.
Case $n \geq 4$.

Take the $n-3$ vertical lines $x=1, x=2, \ldots, x=n-3$.
The remaining points of S_n are those with $a \geq n-2$:

$$W = \{(n-2,1),(n-2,2),(n-2,3),(n-1,1),(n-1,2),(n,1)\}.$$

Cover W by the following three sunny lines:

$$\begin{aligned} L_a &: \text{ through } (n-2,1) \text{ and } (n-1,2) \quad (\text{slope } 1), \\ L_b &: \text{ through } (n-2,2) \text{ and } (n,1) \quad (\text{slope } -\tfrac{1}{2}), \\ L_c &: \text{ through } (n-2,3) \text{ and } (n-1,1) \quad (\text{slope } -2). \end{aligned}$$

All slopes are different from $0, \infty, -1$, hence these lines are sunny.
Together with the $n-3$ vertical lines we have exactly n distinct lines. Every point with $a \leq n-3$ lies on a vertical line; the six points of W lie on L_a, L_b, L_c. Thus the whole S_n is covered. Consequently $k=3$ is attainable for all $n \geq 3$.

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2. Impossibility: no other value of k can occur

We prove by induction on n that any covering of S_n by n lines must have $k \in \{0,1,3\}$.

2.1. The boundary set X and a lemma

Define the three *special* lines

$$S_1 : x=1, \qquad S_2 : y=1, \qquad S_3 : x+y=n+1.$$

These are the sides of the right triangle that contains S_n. Let X be the set of points of S_n lying on these three sides; i.e.

$$X = \{(1,b) \mid 1 \leq b \leq n\} \cup \{(a,1) \mid 1 \leq a \leq n\} \cup \{(a,b) \mid a+b=n+1\}.$$

A1

There are $|X| = 3n-3$ distinct points (the three corners are counted twice)[A2]

Lemma. A line that is **not** one of S_1, S_2, S_3 contains at most two points of X.

Proof. If such a line contained two distinct points from the same side, it would coincide with that side, contradicting the assumption. Hence it can contain at most one point from each side.[A3] Moreover, a straight line can intersect the boundary of a triangle in at most two points (unless it coincides with a side). Therefore it contains at most two points of X. □

2.2. A counting argument for $n \geq 4$

Consider any covering of S_n by n lines, and let k be the number of sunny lines[A4]. For each line ℓ denote by $f(\ell)$ the number of points of X that lie on ℓ. Because every point of X is covered at least once,

$$\sum_{\ell} f(\ell) \geq |X| = 3n-3. \tag{1}$$

If a line is one of the special sides S_1, S_2, S_3, then it contains all n points of that side, so $f(\ell)=n$. If it is not a special side (whether sunny or not), the lemma gives $f(\ell) \leq 2$.
Let s be the number of the three special sides that actually appear among the n lines. Then

$$\sum_{\ell} f(\ell) \leq s \cdot n + (n-s) \cdot 2 = 2n + s(n-2). \tag{2}$$

Combining (1) and (2) we obtain

$$3n-3 \leq 2n + s(n-2) \quad \Longrightarrow \quad n-3 \leq s(n-2). \tag{3}$$

For $n \geq 4$ we have $n-2>0$[A5], so (3) forces $s \geq 1$. Hence **every covering of S_n with $n \geq 4$ lines must contain at least one of the sides $x=1$, $y=1$ or $x+y=n+1$.**

A1 |

Similar to definition of S_n, we need to require a, b are integers.

A2 |

What means of counted twice? Two options: 1. Each edge have n point, so totally $3n$ in 3 edges, but three corners counted twice so $3n-3$ distinct. 2. Directly say $3n-3$ distinct points without mentioning counted twice is also okay.

A3 |

This sentence is not related to the proof.

A4 |

The definition is unnecessary since k is not used in 2.2.

A5 |

$n-3>0$

2.3. Reducing the problem to $n-1$

Assume $n \geq 4$ and that a covering of S_n with n lines and k sunny lines exists. By the previous paragraph, one of the special sides is present; we show how to produce a covering of S_{n-1} with $n-1$ lines and the same k.

Case A: $x = 1$ is present.
Remove the line $x = 1$. The remaining $n-1$ lines cover all points of S_n with $a \geq 2$. Let

$$S' = \{(a, b) \in S_n \mid a \geq 2\},$$

and define the translation $\varphi : S' \to S_{n-1}$ by $\varphi(a, b) = (a-1, b)$[A6]. This is a bijection. Translate the remaining $n-1$ lines by $(-1, 0)$: for each line ℓ (with $\ell \neq (x = 1)$), set $\ell' = \{(x-1, y) \mid (x, y) \in \ell\}$. Translation preserves slopes, so ℓ' is sunny exactly when ℓ is sunny, and distinct lines remain distinct. Because the original lines covered S', their translates cover $\varphi(S') = S_{n-1}$. Thus we obtain a covering of S_{n-1} by $n-1$ lines with exactly k sunny lines.

A6 |
$\phi((a, b))$ since input is a pair (a, b) not two parameter a and b.

Case B: $y = 1$ is present.
Analogously, remove $y = 1$, translate the remaining lines by $(0, -1)$, and obtain a covering of S_{n-1} with $n-1$ lines and k sunny lines.

Case C: $x + y = n + 1$ is present.
Remove this line. The set $\{(a, b) \in S_n \mid a + b \leq n\}$ is exactly S_{n-1}. The remaining $n-1$ lines cover S_{n-1} directly (they cover all of S_n, and the removed line contains no point of S_{n-1}). No translation is needed, and the number of sunny lines stays k.

In all three cases we have produced a covering of S_{n-1} by $n-1$ lines with the same k.

2.4. Base case $n = 3$ and induction

Base case $n = 3$.
We have already exhibited coverings with $k = 0, 1, 3$. It remains to show that $k = 2$ is impossible.

The set S_3 is exactly the boundary X for $n = 3$. Observe that the only lines containing two points of S_3 and being sunny are

$$\begin{aligned} \ell_1 &: \text{ through } (1,1), (2,2) \ (slope 1), \\ \ell_2 &: \text{ through } (1,2), (3,1) \ (slope - 1/2), \\ \ell_3 &: \text{ through } (1,3), (2,1) \ (slope - 2). \end{aligned}$$

Any other line containing two points has slope 0, ∞, or -1, hence is non-sunny.

Suppose a covering of S_3 by three lines has exactly $k = 2$ sunny lines. Then there is exactly one non-sunny line N.

If N is one of the special sides (say $x = 1$), then N covers three points. The remaining three points must be covered by the two sunny lines. But any line through two of those three points is either vertical, horizontal, or of slope -1, all non-sunny. Hence each sunny line can contain at most one of these points, so they cover at most two points - contradiction.

If N is not a special side, then N contains at most two points. Let $f(\ell)$ be the number of points of S_3 on line ℓ. To cover all six points we need $\sum f(\ell) \geq 6$. For a sunny line that is not special, $f(\ell) \leq 2$ (by the lemma). If any line had $f(\ell) \leq 1$, then the total sum would be at most $1 + 2 + 2 = 5 < 6$. Therefore each line must satisfy $f(\ell) = 2$. In particular, the two sunny lines each contain exactly two points, and N also contains exactly two points. Thus the three lines partition S_3 into three disjoint collinear pairs.

The only collinear pairs that can be covered by a sunny line are $\{\ell_1\}, \{\ell_2\}, \{\ell_3\}$. Hence the two sunny lines must be two of $\{\ell_1, \ell_2, \ell_3\}$. The uncovered pair is exactly the pair of the remaining ℓ_m. For N to cover that pair, N would have to be the line through those two points, which is ℓ_m, a sunny line - contradicting that N is non-sunny. Therefore $k = 2$ is impossible. Consequently, for $n = 3$ the only possible k are $0, 1, 3$.

Induction step for $n \geq 4$.
Assume the statement holds for $n-1$. Given any covering of S_n by n lines with k sunny lines, we have shown how to produce a covering of S_{n-1} by $n-1$ lines with the same k. By the induction hypothesis, $k \in \{0, 1, 3\}$. Hence the statement holds for n as well.

By induction, for every $n \geq 3$ any covering of S_n by n lines must have $k \in \{0, 1, 3\}$. Together with the constructions in §1, we conclude that the set of all non-negative integers k for which such a configuration exists is exactly

$$\boxed{\{0,\ 1,\ 3\}}.$$

Human Expert Comment

$$\boxed{7/7}$$

Problem 2

Let Ω and Γ be circles with centres M and N, respectively, such that the radius of Ω is less than the radius of Γ. Suppose Ω and Γ intersect at two distinct points A and B. Line MN intersects Ω at C and Γ at D, so that C, M, N, D lie on MN in that order. Let P be the circumcentre of triangle ACD. Line AP meets Ω again at $E \neq A$ and meets Γ again at $F \neq A$. Let H be the orthocentre of triangle PMN. Prove that the line through H parallel to AP is tangent to the circumcircle of triangle BEF.

Short Answer

No short answer provided.

Model Proof

We set up a coordinate system with $M = (0, 0)$ and $N = (d, 0)$, $d > 0$.
Circles: $\Omega : x^2 + y^2 = r^2$ (radius r), $\Gamma : (x - d)^2 + y^2 = R^2$ (radius R), $0 < r < R$.
They intersect in two points; let $A = (x_A, y_A)$ with $y_A > 0$, $B = (x_A, -y_A)$. Solving the circle equations gives

$$x_A = \frac{d^2 + r^2 - R^2}{2d}, \qquad y_A^2 = \frac{\big((R + r)^2 - d^2\big)\big(d^2 - (R - r)^2\big)}{4d^2}.$$

Define

$$U = R + r, \quad V = R - r, \quad K = d^2 - V^2, \quad L = U^2 - d^2.$$

Then $4y_A^2 = \dfrac{LK}{d^2}$. (Note that d satisfies $V < d < U$ because the circles intersect in two distinct points.)
The line MN (the x-axis) meets Ω at $C = (-r, 0)$ and Γ at $D = (d + R, 0)$. The order on MN is C, M, N, D.

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1. Circumcenter P of $\triangle ACD$

Since C and D lie on the x-axis, the perpendicular bisector of CD is the vertical line $x = h$ where

$$h = \frac{-r + (d + R)}{2} = \frac{d + R - r}{2} = \frac{d + V}{2}.$$

Thus $P = (h, p)$ for some p. From $PA = PC$ we obtain

$$(h - x_A)^2 + (p - y_A)^2 = (h + r)^2 + p^2.$$

Solving for p yields

$$p = -\frac{K\,T}{4d\,y_A}, \qquad \text{where } T = d + R + r = d + U.$$

The vector $\mathbf{v} = P - A$ is

$$\mathbf{v} = (v_x, v_y) = \left(\frac{VT}{2d},\ -\frac{KUT}{4d^2 y_A}\right).$$

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2. Second intersections E, F and midpoint S of EF

The line AP consists of points $A + t\mathbf{v}$. Substituting into Ω (using $A \cdot A = r^2$) gives

$$2t\,A \cdot \mathbf{v} + t^2|\mathbf{v}|^2 = 0,$$

so the non-zero root is $t_E = -2A \cdot \mathbf{v}/|\mathbf{v}|^2$. Compute

$$A \cdot \mathbf{v} = -\frac{rT}{2} \quad \Longrightarrow \quad t_E = \frac{rT}{|\mathbf{v}|^2}.$$

Similarly, for Γ we use $(A - N) \cdot \mathbf{v} = -RT/2$ and obtain

$$t_F = \frac{RT}{|\mathbf{v}|^2}.$$

Hence $E = A + t_E\mathbf{v},\ F = A + t_F\mathbf{v}$. The midpoint S of EF is $S = A + t_S\mathbf{v}$ with

$$t_S = \frac{t_E + t_F}{2} = \frac{(r+R)T}{2|\mathbf{v}|^2} = \frac{UT}{2|\mathbf{v}|^2}.$$

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3. Unit vectors along and perpendicular to AP

Let $\mathbf{e} = \mathbf{v}/|\mathbf{v}|$ and $\mathbf{n} = (-v_y, v_x)/|\mathbf{v}|$. Then $\mathbf{e}$ is the direction of AP, $\mathbf{n}$ is perpendicular to AP, and EF is parallel to $\mathbf{e}$. The half-length of EF is

$$\ell = \frac{|t_F - t_E|\,|\mathbf{v}|}{2} = \frac{VT}{2|\mathbf{v}|}, \qquad \ell^2 = \frac{V^2T^2}{4|\mathbf{v}|^2}.$$

Thus $E = S - \ell\mathbf{e},\ F = S + \ell\mathbf{e}$.

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4. Orthocenter H of $\triangle PMN$

Points: $M = (0,0),\ N = (d,0),\ P = (h,p)$.
The altitude from P is the vertical line $x = h$.
The altitude from M is perpendicular to PN; slope of PN is $-p/(d-h)$, so the altitude from M has equation $y = \frac{d-h}{p}x$.
Intersection gives

$$H = \left(h,\ \frac{h(d-h)}{p}\right).$$

Now $h(d-h) = \frac{(d+V)(d-V)}{4} = \frac{d^2-V^2}{4} = \frac{K}{4}$. Using $p = -\frac{KT}{4dy_A}$, we obtain

$$H = \left(h,\ \frac{K/4}{-KT/(4dy_A)}\right) = \left(h,\ -\frac{dy_A}{T}\right).$$

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5. Computation of $\beta = (S-B)\cdot\mathbf{n}$ and $\gamma = (S-H)\cdot\mathbf{n}$

Because $S = A + t_S\mathbf{v}$ and $\mathbf{n} \perp \mathbf{v}$, we have

$$(S-B)\cdot\mathbf{n} = (A-B)\cdot\mathbf{n}, \qquad (S-H)\cdot\mathbf{n} = (A-H)\cdot\mathbf{n}.$$

Thus

$$\beta = (A-B)\cdot\mathbf{n}, \qquad \gamma = (A-H)\cdot\mathbf{n}.$$

Now $A - B = (0, 2y_A)$ and $\mathbf{n} = (-v_y, v_x)/|\mathbf{v}|$, so

$$\beta = \frac{2y_Av_x}{|\mathbf{v}|} = \frac{2y_A}{|\mathbf{v}|}\cdot\frac{VT}{2d} = \frac{y_AVT}{d\,|\mathbf{v}|}.$$

To compute γ, we use $\mathbf{w}\cdot\mathbf{n} = -\frac{\mathbf{w}\times\mathbf{v}}{|\mathbf{v}|}$ (for any $\mathbf{w}$). Hence

$$\gamma = -\frac{(A-H)\times\mathbf{v}}{|\mathbf{v}|}.$$

We compute the cross product. Coordinates:

$$x_A - h = -\frac{VT}{2d}, \qquad y_A - H_y = y_A + \frac{dy_A}{T} = y_A\frac{T+d}{T},$$
$$v_x = \frac{VT}{2d}, \qquad v_y = -\frac{KUT}{4d^2y_A}.$$

Then

$$(A-H)\times\mathbf{v} = (x_A-h)v_y - (y_A-H_y)v_x = \left(-\frac{VT}{2d}\right)\left(-\frac{KUT}{4d^2y_A}\right) - \left(y_A\frac{T+d}{T}\right)\left(\frac{VT}{2d}\right) = \frac{VKUT^2}{8d^3y_A} - \frac{Vy_A(T+d)}{2d}.$$

Factor $\frac{V}{2d}$:

$$(A-H)\times\mathbf{v}=\frac{V}{2d}\left(\frac{KUT^2}{4d^2y_A}-y_A(T+d)\right).$$

Let $B=\frac{KUT^2}{4d^2y_A}-y_A(T+d)$. Multiply by $2y_A$:

$$2y_AB=\frac{KUT^2}{2d^2}-2y_A^2(T+d).$$

Using $4y_A^2=\frac{LK}{d^2}$ we have $y_A^2=\frac{LK}{4d^2}$, so

$$2y_A^2(T+d)=\frac{LK(T+d)}{2d^2}.$$

Thus

$$2y_AB=\frac{K}{2d^2}\big(UT^2-L(T+d)\big).$$

Now $T=d+U$ and $L=U^2-d^2$. Compute

$$UT^2-L(T+d)=U(d+U)^2-(U^2-d^2)(2d+U)=2d^2T.$$

Hence $2y_AB=\frac{K}{2d^2}\cdot 2d^2T=KT$, so $B=\frac{KT}{2y_A}$. Consequently,

$$(A-H)\times\mathbf{v}=\frac{V}{2d}\cdot\frac{KT}{2y_A}=\frac{VKT}{4dy_A}.$$

Therefore

$$\gamma=-\frac{VKT}{4dy_A|\mathbf{v}|}.$$

Comparing with $\beta=\dfrac{y_AVT}{d|\mathbf{v}|}$, we obtain

$$\gamma=-\frac{K}{4y_A^2}\,\beta.$$

Moreover, using $4y_A^2=\dfrac{LK}{d^2}$ we can write

$$\frac{K}{4y_A^2}=\frac{d^2}{L}\quad\Longrightarrow\quad\gamma=-\frac{d^2}{L}\,\beta.$$

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6. Expression for $|S-B|^2$

Write $A-B=\alpha\mathbf{e}+\beta\mathbf{n}$, where $\alpha=(A-B)\cdot\mathbf{e}=\dfrac{(A-B)\cdot\mathbf{v}}{|\mathbf{v}|}$. Compute

$$(A-B)\cdot\mathbf{v}=2y_Av_y=2y_A\left(-\frac{KUT}{4d^2y_A}\right)=-\frac{KUT}{2d^2},$$

so

$$\alpha=-\frac{KUT}{2d^2|\mathbf{v}|}.$$

Since $S=A+t_S\mathbf{v}$ and $\mathbf{v}=|\mathbf{v}|\mathbf{e}$,

$$S-B=(A-B)+t_S\mathbf{v}=(\alpha\mathbf{e}+\beta\mathbf{n})+t_S|\mathbf{v}|\mathbf{e}=(\alpha+t_S|\mathbf{v}|)\mathbf{e}+\beta\mathbf{n}.$$

Now $t_S|\mathbf{v}|=\dfrac{UT}{2|\mathbf{v}|}$. Thus

$$\alpha+t_S|\mathbf{v}|=-\frac{KUT}{2d^2|\mathbf{v}|}+\frac{UT}{2|\mathbf{v}|}=\frac{UT}{2|\mathbf{v}|}\left(1-\frac{K}{d^2}\right)=\frac{UT}{2|\mathbf{v}|}\cdot\frac{V^2}{d^2}=\frac{UV^2T}{2d^2|\mathbf{v}|}.$$

Hence

$$|S-B|^2 = (\alpha + t_S|\mathbf{v}|)^2 + \beta^2 = \frac{U^2V^4T^2}{4d^4|\mathbf{v}|^2} + \frac{y_A^2V^2T^2}{d^2|\mathbf{v}|^2}.$$

Substitute $y_A^2 = \dfrac{LK}{4d^2}$ to obtain

$$|S-B|^2 = \frac{V^2T^2}{|\mathbf{v}|^2}\left(\frac{U^2V^2}{4d^4} + \frac{LK}{4d^4}\right) = \frac{V^2T^2}{4d^4|\mathbf{v}|^2}\big(U^2V^2 + LK\big).$$

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7. Circumcenter O of $\triangle BEF$ and the tangency condition

The perpendicular bisector of EF is the line through S parallel to $\mathbf{n}$, so we can write $O = S + \lambda\mathbf{n}$ for some real λ. The circumradius ρ satisfies $\rho^2 = |O-E|^2 = \ell^2 + \lambda^2$. The condition $|O-B|^2 = \rho^2$ gives

$$|(S-B) + \lambda\mathbf{n}|^2 = \ell^2 + \lambda^2.$$

Expanding, and using $(S-B) = \alpha'\mathbf{e} + \beta\mathbf{n}$ with $\alpha' = \alpha + t_S|\mathbf{v}|$ (so that $|S-B|^2 = \alpha'^2 + \beta^2$), we get

$$\alpha'^2 + (\beta+\lambda)^2 = \ell^2 + \lambda^2 \implies |S-B|^2 + 2\lambda\beta = \ell^2. \tag{1}$$

Thus $2\lambda\beta = \ell^2 - |S-B|^2$.

Now consider the line through H parallel to AP. Its distance to O equals the circumradius of $\triangle BEF$ iff

$$\big|(O-H)\cdot\mathbf{n}\big| = \rho.$$

Since $O - H = (S-H) + \lambda\mathbf{n}$ and $(S-H)\cdot\mathbf{n} = \gamma$, we have $(O-H)\cdot\mathbf{n} = \gamma + \lambda$. Therefore tangency is equivalent to

$$(\gamma+\lambda)^2 = \lambda^2 + \ell^2 \iff \gamma^2 + 2\gamma\lambda = \ell^2. \tag{2}$$

Substituting $\lambda = \dfrac{\ell^2 - |S-B|^2}{2\beta}$ from (1) into (2) yields

$$\gamma^2 + \frac{\gamma}{\beta}\big(\ell^2 - |S-B|^2\big) = \ell^2.$$

Multiplying by β and rearranging gives

$$\gamma\big(\ell^2 - |S-B|^2\big) = \beta\big(\ell^2 - \gamma^2\big). \tag{3}$$

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8. Verification of (3)

We now use the explicit expressions:

$$\beta = \frac{y_AVT}{d\,|\mathbf{v}|}, \quad \gamma = -\frac{d^2}{L}\,\beta, \quad \ell^2 = \frac{V^2T^2}{4|\mathbf{v}|^2}, \quad |S-B|^2 = \frac{V^2T^2}{4d^4|\mathbf{v}|^2}\big(U^2V^2 + LK\big).$$

Set $c = -\dfrac{d^2}{L}$, so that $\gamma = c\beta$. Then (3) becomes (after dividing by the non-zero factor β)

$$c\big(\ell^2 - |S-B|^2\big) = \ell^2 - c^2\beta^2. \tag{3'}$$

Compute each side:

$$c^2\beta^2 = \frac{d^4}{L^2}\cdot\frac{y_A^2V^2T^2}{d^2|\mathbf{v}|^2} = \frac{d^4}{L^2}\cdot\frac{LKV^2T^2}{4d^4|\mathbf{v}|^2} = \frac{KV^2T^2}{4L|\mathbf{v}|^2}.$$

Hence

$$\ell^2 - c^2\beta^2 = \frac{V^2T^2}{4|\mathbf{v}|^2} - \frac{KV^2T^2}{4L|\mathbf{v}|^2} = \frac{V^2T^2}{4|\mathbf{v}|^2}\left(1 - \frac{K}{L}\right) = \frac{V^2T^2}{4|\mathbf{v}|^2}\cdot\frac{L-K}{L}.$$

Next,

$$\ell^2 - |S-B|^2 = \frac{V^2T^2}{4|\mathbf{v}|^2} - \frac{V^2T^2}{4d^4|\mathbf{v}|^2}\big(U^2V^2 + LK\big) = \frac{V^2T^2}{4|\mathbf{v}|^2}\left[1 - \frac{U^2V^2+LK}{d^4}\right].$$

Thus the left-hand side of (3') is

$$c\big(\ell^2 - |S-B|^2\big) = -\frac{d^2}{L}\cdot\frac{V^2T^2}{4|\mathbf{v}|^2}\left[1 - \frac{U^2V^2+LK}{d^4}\right] = -\frac{V^2T^2}{4L|\mathbf{v}|^2}\left(d^2 - \frac{U^2V^2+LK}{d^2}\right).$$

Equation (3') is therefore equivalent (after multiplying both sides by $4L|\mathbf{v}|^2/V^2T^2$) to

$$-d^2 + \frac{U^2V^2+LK}{d^2} = L - K.$$

Multiply by d^2:

$$-d^4 + U^2V^2 + LK = (L-K)d^2.$$

Now substitute $L = U^2 - d^2$, $K = d^2 - V^2$. The left side expands to

$$-d^4 + U^2V^2 + (U^2-d^2)(d^2-V^2) = -d^4 + U^2V^2 + U^2d^2 - U^2V^2 - d^4 + d^2V^2 = d^2(U^2+V^2-2d^2).$$

The right side is

$$(L-K)d^2 = \big((U^2-d^2) - (d^2-V^2)\big)d^2 = (U^2+V^2-2d^2)d^2,$$

which is identical. Hence (3') holds, and consequently (3) and (2) are satisfied.
Therefore the line through H parallel to AP is tangent to the circumcircle of $\triangle BEF$. $\square$

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Input Your input will consist of: * **Problem Statement**: A mathematical problem that the proof is attempting to solve.
* **Reference Solution**: A correct solution or proof provided for reference. This is **not necessarily the only valid solution**. If the problem requires a final numeric or algebraic answer, this section contains the correct answer, which should be the only accepted final answer (though alternative reasoning paths are valid).
* **Marking Scheme**: A problem-specific grading rubric (0-7 scale) with checkpoints, zero-credit items, and deductions. You must follow this scheme when assigning points.
* **Proof Solution**: The proof that you need to evaluate. This proof may contain errors, omissions, or unclear steps. The proof was generated by another language model.
Task Analyze the proof carefully.
* Compare the proof against the reference solution and the marking scheme.
* Award points according to the marking scheme's checkpoints, zero-credit items, and deductions.
* Identify logical errors, incorrect steps, or unclear reasoning.
* Give a score between 0 and 7 with a brief overall assessment.
* Show clearly how the score was derived:
* Which checkpoints were earned (with awarded points).
Grade this as an IMO marker using ultra-strict standards. Do not fill in missing steps, do not repair the proof, and do not infer claims that are not explicitly proved. Treat any later use of an unstated or unreproved bound as a gap. First list every logical dependency line by line, then check whether each dependency has been established earlier in the written solution. Distinguish clearly between: (1) correct idea, (2) complete written proof, and (3) patchable gap. Score according to the written proof, not the intended argument
problem: {problem}
Reference Solution: {reference solution}

Marking Scheme: {marking scheme}
proof: {proof}

GPT-5.4-Thinking(Extensive)

Thought for 2m 35s>

Here is the grading, done as an ultra-strict IMO marker on the written proof only.
Overall judgment: the submitted proof is mathematically sound and, despite being very calculation-heavy, it is a complete coordinate proof. It does not follow the reference solution's synthetic route, but it replaces the official constructions by explicit coordinate/algebraic identities and closes the tangency criterion directly. I do not see a fatal logical gap.
Dependency audit
1. The proof fixes coordinates (M=(0,0)), (N=(d,0)), with circles

$$\Omega : x^2 + y^2 = r^2, \qquad \Gamma : (x-d)^2 + y^2 = R^2, \qquad 0 < r < R.$$

This is a valid coordinate setup. No gap.
2. It writes

$$A = (x_A, y_A), \qquad B = (x_A, -y_A),$$

and derives

$$x_A = \frac{d^2 + r^2 - R^2}{2d}, \qquad y_A^2 = \frac{((R+r)^2 - d^2)(d^2 - (R-r)^2)}{4d^2}.$$

This follows from solving the two circle equations and using symmetry about the (x)-axis. Established.
3. It defines

$$U = R + r, \quad V = R - r, \quad K = d^2 - V^2, \quad L = U^2 - d^2,$$

and notes (V<d<U) because the circles intersect in two distinct points. Hence $(K > 0), (L > 0), (y_A > 0)$. Established.
4. It identifies

$$C = (-r, 0), \qquad D = (d + R, 0),$$

with order (C,M,N,D). Correct and established.
5. It finds the circumcenter (P) of (△ ACD) as (P=(h,p)), where

$$h = \frac{-r + (d+R)}{2} = \frac{d+V}{2}.$$

This uses that the perpendicular bisector of (CD) is vertical. Established.
6. From (PA=PC), it states

$$p = -\frac{KT}{4dy_A}, \qquad T = d + U.$$

This is a compressed algebra step, but it is a direct consequence of the displayed equation and the earlier definitions. I checked it; it is correct. No logical gap.
7. It defines ($\mathbf{v} = P - A$) and gives

$$\mathbf{v} = (v_x, v_y) = \left(\frac{VT}{2d}, -\frac{KUT}{4d^2 y_A}\right).$$

Again, compressed algebra, but correct and dependent only on already established formulas.
8. It parametrizes line (AP) as $(A + t\mathbf{v})$. Substituting into (Ω), it derives the second intersection parameter

$$t_E = \frac{rT}{|\mathbf{v}|^2}.$$

This follows from

$$2t, A \cdot \mathbf{v} + t^2|\mathbf{v}|^2 = 0$$

and the computed identity ($A \cdot \mathbf{v} = -rT/2$). Established.
9. Similarly, substituting into (Γ), it derives

$$t_F = \frac{RT}{|\mathbf{v}|^2}.$$

This depends on ($(A - N) \cdot \mathbf{v} = -RT/2$). Established.
10. Hence

$$E = A + t_E\mathbf{v}, \qquad F = A + t_F\mathbf{v},$$

and the midpoint (S) of (EF) is

$$S = A + t_S\mathbf{v}, \qquad t_S = \frac{UT}{2|\mathbf{v}|^2}.$$

Established.
11. It defines

$$\mathbf{e} = \frac{\mathbf{v}}{|\mathbf{v}|}, \qquad \mathbf{n} = \frac{(-v_y, v_x)}{|\mathbf{v}|},$$

so ($\mathbf{e}$) is the direction of (AP), ($\mathbf{n} \perp AP$), and ($EF \parallel AP$). Established.
12. It computes the half-length of (EF):

$$\ell = \frac{|t_F - t_E|, |\mathbf{v}|}{2} = \frac{VT}{2|\mathbf{v}|}.$$

Correct and established.
13. It computes the orthocenter (H) of ($\triangle PMN$). From the altitude equations it gets

$$H = \left(h, \frac{h(d-h)}{p}\right),$$

and then

$$H = \left(h, -\frac{dy_A}{T}\right).$$

This is correct and fully supported by earlier formulas.
14. It defines

$$\beta = (S - B) \cdot \mathbf{n}, \qquad \gamma = (S - H) \cdot \mathbf{n}.$$

Using ($S = A + t_S\mathbf{v}$) and ($\mathbf{n} \perp \mathbf{v}$), it reduces these to

$$\beta = (A - B) \cdot \mathbf{n}, \qquad \gamma = (A - H) \cdot \mathbf{n}.$$

Established.
15. It computes

$$\beta = \frac{y_A VT}{d|\mathbf{v}|}.$$

Correct and established.
16. It computes

$$\gamma = -\frac{(A - H) \times \mathbf{v}}{|\mathbf{v}|},$$

then works through the cross product and arrives at

$$\gamma = -\frac{VKT}{4dy_A|\mathbf{v}|} = -\frac{K}{4y_A^2}\beta = -\frac{d^2}{L}\beta.$$

This is the densest calculation in the proof, but every ingredient used has already been introduced, and the final identity checks out. No missing geometric lemma is being smuggled in.
17. It decomposes (A-B) into the ($(\mathbf{e}, \mathbf{n})$)-basis, then obtains

$$|S - B|^2 = \frac{V^2T^2}{4d^4|\mathbf{v}|^2}\left(U^2V^2 + LK\right).$$

This is a long but direct algebraic consequence of earlier formulas. Established.

18. It introduces (O), the circumcenter of ($\triangle BEF$). Since ($EF \parallel \mathbf{e}$), the perpendicular bisector of (EF) is the line through (S) parallel to ($\mathbf{n}$), so

$$O = S + \lambda \mathbf{n}.$$

Correct.
19. Using ($|OB| = |OE|$), it derives

$$|S - B|^2 + 2\lambda\beta = \ell^2. \tag{1}$$

Correct.
20. It then expresses the line through (H) parallel to (AP) as a line with direction ($\mathbf{e}$), so its distance from (O) is ($|(O - H) \cdot \mathbf{n}|$). Tangency is therefore equivalent to

$$(\gamma + \lambda)^2 = \lambda^2 + \ell^2 \quad \Longleftrightarrow \quad \gamma^2 + 2\gamma\lambda = \ell^2. \tag{2}$$

This is a valid tangent-to-circle criterion.
21. Eliminating (λ) between (1) and (2), it obtains

$$\gamma(\ell^2 - |S - B|^2) = \beta(\ell^2 - \gamma^2). \tag{3}$$

Correct algebra.
22. It substitutes the explicit expressions

$$\beta,\ \gamma,\ \ell^2,\ |S - B|^2$$

into (3), rewrites with ($c = -d^2/L$), and reduces the whole claim to

$$-d^4 + U^2V^2 + LK = (L - K)d^2.$$

Correct reduction.
23. Finally, substituting ($L = U^2 - d^2$) and ($K = d^2 - V^2$), both sides simplify to

$$d^2(U^2 + V^2 - 2d^2),$$

so (3) holds, hence (2) holds, hence the line through (H) parallel to (AP) is tangent to the circumcircle of ($\triangle BEF$). Complete.
Correct idea vs complete proof vs patchable gaps
Correct idea: yes. The idea is to replace the synthetic construction by a coordinate model, compute (P,E,F,H), write the circumcenter (O) of ($\triangle BEF$) in the (($\mathbf{e}, \mathbf{n}$))-frame, and reduce tangency to an explicit scalar identity.
Complete written proof: yes. Every major object used later is explicitly defined, and the final tangency condition is actually proved, not merely asserted.
Patchable gaps: there are a few compressed algebra jumps ("solving for (p)", the dot-product simplifications, the cross-product simplification, the final reduction in Section 8), but these are routine algebraic consequences of already displayed formulas, not new unproved geometric claims. Under IMO marking, I would not count these as logical gaps.
Checkpoint scoring
Because this is a full alternative coordinate proof, I map the official checkpoints by equivalent accomplished work rather than by the exact synthetic route.
1. Foundational Angle Chase — 1/1 Earned by equivalent foundational setup: the proof correctly derives the full coordinate description of (A,B,C,D,P), including the intersection formulas and all sign/positivity conditions needed later.
2. Parallelogram Construction and Alignment — 2/2 The proof does not use the official (A')-parallelogram route, but it establishes the equivalent control of the (E,F) configuration directly: it computes the line (AP), the second intersections (E,F), the midpoint (S), and the direction/normal frame (($\mathbf{e}, \mathbf{n}$)). This is enough to replace the official CP2 machinery.
3. Properties of Point (T) — 2/2 The proof bypasses the auxiliary point (T) entirely and instead analyzes the actual circumcenter (O) of ($\triangle BEF$) directly. This is a valid stronger substitute for the official CP3 structure.

4. Properties of (H) and Final Conclusion — 2/2 Earned. The proof computes (H) explicitly, computes its signed normal offset (γ), and then verifies the tangency criterion exactly.
Deductions
No cap applies, because the proof reaches the final tangency statement.
No (-1) deduction applies. I do not see a nontrivial unproved leap in the main line.
Final score

$$\boxed{7/7}$$

Brief assessment
This is a valid full-score coordinate solution. It is not elegant, and it is denser than the reference synthetic proof, but as written it does close all the required dependencies. Under ultra-strict marking, I would still award full marks because the proof is complete rather than merely suggestive.

Problem 3

Let $\mathbb{N}$ denote the set of positive integers. A function $f : \mathbb{N} \to \mathbb{N}$ is said to be bonza if

$$f(a) \text{ divides } b^a - f(b)^{f(a)}$$

for all positive integers a and b.
Determine the smallest real constant c such that $f(n) \leq cn$ for all bonza functions f and all positive integers n.

Short Answer
4

Model Proof

We will prove that the smallest constant c such that $f(n) \leq cn$ for every bonza function f is $c = 4$.

1. Basic properties

From the definition, for all $a, b \in \mathbb{N}$,

$$f(a) \mid b^a - f(b)^{f(a)}. \tag{*}$$

- Taking $b = a$ gives $f(a) \mid a^a - f(a)^{f(a)}$. Since $f(a)^{f(a)}$ is a multiple of $f(a)$, we obtain $f(a) \mid a^a$. Hence every prime divisor of $f(a)$ divides a.
- Taking $a = b = 1$ yields $f(1) \mid 1 - f(1)^{f(1)}$. The right-hand side is congruent to 1 modulo $f(1)$, so $f(1) \mid 1$. Thus $f(1) = 1$.

2. Behaviour on primes

Let p be a prime. Because every prime divisor of $f(p)$ must divide p, we have $f(p) = p^t$ for some integer $t \geq 0$ (where $t = 0$ means $f(p) = 1$). Define

$$S = \{\, p \text{ prime} \mid f(p) \neq 1 \,\} = \{\, p \mid t \geq 1 \,\}.$$

B1

B1 |

t should be a function w.r.t p, like $t(p)$ or t_p, instead of a constant

If $p \in S$, then $f(p) = p^t$ with $t \geq 1$. Using (*) with $a = p$ and an arbitrary b we get

$$p^t \mid b^p - f(b)^{p^t}.$$

In particular, reducing modulo p, we have

$$b^p \equiv f(b)^{p^t} \pmod{p}.$$

We now show that $x^{p^t} \equiv x \pmod{p}$ for every integer x. This follows by induction on t using Fermat's little theorem: for $t = 0$ it is trivial; assuming $x^{p^t} \equiv x \pmod{p}$, then

$$x^{p^{t+1}} = (x^{p^t})^p \equiv x^p \pmod{p} \equiv x \pmod{p},$$

where the last congruence is Fermat's little theorem. Hence $f(b)^{p^t} \equiv f(b) \pmod{p}$. Also $b^p \equiv b \pmod{p}$ by Fermat. Therefore

$$b - f(b) \equiv b^p - f(b)^{p^t} \equiv 0 \pmod{p},$$

so

$$f(b) \equiv b \pmod{p} \qquad \text{for all } b \in \mathbb{N}. \tag{1}$$

3. When S is infinite

Assume that S is infinite. Then for any fixed n, the integer $f(n) - n$ is divisible by every prime in S. An integer that is divisible by infinitely many distinct primes must be zero. Hence $f(n) = n$ for all n. Consequently, the only bonza function with S infinite is the identity $f(n) = n$.

4. Odd primes in S force S to be infinite

Suppose that an odd prime p belongs to S. Let q be any prime with $q \not\equiv 1 \pmod p$. From (1) we have $f(q) \equiv q \pmod p$. Because $f(q) \mid q^q$, the only prime divisors of $f(q)$ are q; thus $f(q)$ is a power of q, say $f(q) = q^s$. If $s = 0$ then $f(q) = 1$, which would imply $q \equiv 1 \pmod p$, contradicting the choice of q. Hence $s \geq 1$, and therefore $q \mid f(q)$, so $q \in S$.

We now prove that there are infinitely many primes q with $q \not\equiv 1 \pmod p$. Assume, to the contrary, that only finitely many such primes exist; list them as $q_1, q_2, \ldots, q_k$ (this list contains p and 2 because both are not $\equiv 1 \pmod p$). Consider

$$N = p \cdot (q_1 q_2 \cdots q_k) - 1.$$

B2 Clearly $N > 1$. Any prime divisor r of N is different from every q_i, for otherwise r would divide $p \cdot \prod q_i$ and hence divide $N + 1$, forcing $r \mid 1$, impossible. By our assumption, every prime divisor of N satisfies $r \equiv 1 \pmod p$. Consequently, $N \equiv 1 \pmod p$. But $N = p \cdot \prod q_i - 1 \equiv -1 \pmod p$, a contradiction. Hence there are infinitely many primes q with $q \not\equiv 1 \pmod p$. By the previous paragraph, each such prime lies in S, so S is infinite. Then by §3, f is the identity.

B2 |

correct, but no need to have a multiply as p here, since you already said p and 2 in the list.

Thus, if f is not the identity, the set S cannot contain any odd prime. Therefore the only possibilities for a non-identity bonza function are

$$\text{(i) } S = \varnothing \qquad \text{or} \qquad \text{(ii) } S = \{2\}.$$

5. Case $S = \varnothing$

This means $f(p) = 1$ for every prime p. We show that then f is the constant function 1.

Suppose, for contradiction, that there exists a with $f(a) > 1$. Let r be a prime divisor of $f(a)$. From §1 we have $r \mid a$. Since r is prime and $S = \varnothing$, we have $f(r) = 1$. Apply (*) with a (the original number) and $b = r$:

$$f(a) \mid r^a - f(r)^{f(a)} = r^a - 1.$$

Because $r \mid f(a)$, we obtain $r \mid r^a - 1$. But $r^a \equiv 0 \pmod r$, so $r^a - 1 \equiv -1 \pmod r$, contradiction. Hence no such a exists, and $f(n) = 1$ for all n. The constant function 1 is clearly bonza because 1 divides any integer.

6. Case $S = \{2\}$

Here $f(2) \neq 1$ (and $f(2) \mid 2^2 = 4$, so $f(2) = 2$ or 4), while $f(p) = 1$ for every odd prime p. We will prove that $f(n) \leq 4n$ for all n.

6.1 Odd arguments

Let $n > 1$ be odd. Assume $f(n) > 1$ and let r be a prime divisor of $f(n)$. Then $r \mid n$ (since $f(n) \mid n^n$), so r is an odd prime. Because r is odd, $f(r) = 1$. Using (*) with $a = n$ and $b = r$ gives

$$f(n) \mid r^n - 1.$$

But $r \mid f(n)$ implies $r \mid r^n - 1$, which is impossible because $r^n \equiv 0 \pmod r$. Therefore $f(n) = 1$ for all odd n. Also $f(1) = 1$.[B3]

B3 |

Logical issue. It should be "Therefore, $f(n) = 1$ for odd $n > 1$. Also $f(1) = 1$, so $f(\text{odd}) = 1$".

6.2 Even arguments

Now let n be even and write $n = 2^k \cdot m$ with $k = v_2(n) \geq 1$ and m odd.

First we show that $f(n)$ is a power of 2. Suppose an odd prime r divides $f(n)$. Then, as in §6.1, $r \mid n$ and $f(r) = 1$ (since r is odd), and (*) with $b = r$[B4] yields $f(n) \mid r^n - 1$, leading to the same contradiction $r \mid r^n - 1$. Hence all prime factors of $f(n)$ are 2, so $f(n) = 2^s$ for some $s \geq 0$.

B4 |

missing $a = n$

Now take b to be any odd prime p[B5]. Because $f(p) = 1$, condition (*) gives

B5 |

missing $a = n$

$$2^s \mid p^n - 1. \tag{2}$$

Thus $s \le v_2(p^n - 1)$ for every odd prime p. Consequently,

$$s \le \min_{\text{odd prime } p} v_2(p^n - 1).$$

We compute this minimum. Since n is even, write $n = 2^k \cdot m$ with m odd[B6]. For an odd prime p, we have (using the lifting-the-exponent lemma or a direct argument)

$$v_2(p^n - 1) = v_2(p^{2^k} - 1) = v_2(p-1) + v_2(p+1) + k - 1. \tag{3}$$

Proof of (3): Because m is odd, $v_2(p^n - 1) = v_2(p^{2^k} - 1)$. For $k \ge 1$, one proves by induction that $v_2(p^{2^k} - 1) = v_2(p-1) + v_2(p+1) + k - 1$. The base $k = 1$ is $p^2 - 1 = (p-1)(p+1)$. For the inductive step, $p^{2^k} - 1 = (p^{2^{k-1}} - 1)(p^{2^{k-1}} + 1)$. The first factor contributes $v_2(p^{2^{k-1}} - 1)$, and for $k \ge 2$ one has $v_2(p^{2^{k-1}} + 1) = 1$ because $p^{2^{k-1}} \equiv 1 \pmod 8$ (as p odd implies $p^2 \equiv 1 \pmod 8$, and by induction all higher powers are $\equiv 1 \pmod 8$). This gives the formula.

For an odd prime p, the two consecutive even numbers $p - 1$ and $p + 1$ have the following property: one of them is divisible by 2 exactly once, the other is divisible by 4 (at least). Hence

$$v_2(p-1) + v_2(p+1) \ge 1 + 2 = 3,$$

and the minimum 3 is attained for infinitely many primes, e.g. $p = 3, 5, 11, 13, \ldots$. Therefore

$$\min_{\text{odd prime } p} v_2(p^n - 1) = 3 + k - 1 = k + 2.$$

Thus $s \le k + 2$, and

$$f(n) = 2^s \le 2^{k+2} = 4 \cdot 2^k.$$

Since $n = 2^k \cdot m$ with $m \ge 1$, we have $2^k \le n$. Hence

$$f(n) \le 4 \cdot 2^k \le 4n.$$

For odd n we already have $f(n) = 1 \le 4n$. The identity function $f(n) = n$ also satisfies this bound. Consequently, **every** bonza function satisfies $f(n) \le 4n$ for all n.[B7]

B6 |

already defined at the beginning of 6.2

B7 |

You divided into 3 cases, and here ignored the situation with $f(n) = 1$ always.

7. Sharpness

We construct a bonza function F for which $F(4) = 16$, showing that c cannot be smaller than 4.

Define

$$F(1) = 1, \qquad F(2) = 2,$$
$$F(n) = 1 \quad \text{for odd } n > 1,$$

and for even $n \ge 4$, let $k = v_2(n)$ and set $F(n) = 4 \cdot 2^k = 2^{k+2}$.

(One could also take $F(2) = 4$; both work, but we choose $F(2) = 2$ for definiteness.)

We verify that F satisfies the bonza condition.

- **Case a odd:** Then $F(a) = 1$ and $1 \mid b^a - F(b)^1$ trivially.
- **Case $a = 2$:**
  b odd: $F(2) = 2$ divides $b^2 - 1$ because b^2 is odd, so $b^2 - 1$ is even.
  b even: Both b^2 and $F(b)^2$ are even, so their difference is even, hence divisible by 2.
- **Case $a \ge 4$ even:** Write $a = 2^k \cdot m$ with $k = v_2(a) \ge 1$ and m odd, so $F(a) = 2^{k+2}$. Consider b.
  b odd: $F(b) = 1$. For any odd b and even a, we have
  $$v_2(b^a - 1) = v_2(b-1) + v_2(b+1) + k - 1 \ge 3 + k - 1 = k + 2,$$
  so $2^{k+2} \mid b^a - 1$.
  b even: Write $F(b) = 2^t$ with $t \ge 1$ (if $b = 2$ then $t = 1$; if $b \ge 4$ then $t = v_2(b) + 2 \ge 3$).

Then

$$v_2(b^a) = a \cdot v_2(b) \geq a \geq k+2 \quad \text{(since } a \geq 4 \text{ and even)},$$

and

$$v_2\big(F(b)^{F(a)}\big) = t \cdot F(a) = t \cdot 2^{k+2} \geq 2^{k+2}.$$

B8 Hence both b^a and $F(b)^{F(a)}$ are multiples of 2^{k+2}, and therefore their difference is also a multiple of 2^{k+2}.

B8 |
$k+2$ should be enough

Thus in all situations $F(a) \mid b^a - F(b)^{F(a)}$. Therefore F is bonza. For $n = 4$ we have $v_2(4) = 2$, so $F(4) = 2^4 = 16$, giving $F(4)/4 = 4$. Hence any constant c with $f(n) \leq cn$ for all bonza functions must satisfy $c \geq 4$.

8. Conclusion

We have shown that every bonza function f satisfies $f(n) \leq 4n$ for all $n \in \mathbb{N}$, and we have exhibited a bonza function for which equality $f(4) = 4 \cdot 4$ is attained (i.e., $f(4)/4 = 4$). Consequently, the smallest possible constant is

$$\boxed{4}.$$

Human Expert Comment

$\boxed{7/7}$

Problem 4

A proper divisor of a positive integer N is a positive divisor of N other than N itself.
The infinite sequence $a_1, a_2, \cdots$ consists of positive integers, each of which has at least three proper divisors.
For each $n \geq 1$, the integer a_{n+1} is the sum of the three largest proper divisors of a_n.
Determine all possible values of a_1.

Short Answer
The set of all possible values for a_1 are integers a of the form $a = 2^x 3^y m$, where: 1. x is a positive odd integer. 2. y is a positive integer satisfying $y \geq \frac{x+1}{2}$. 3. m is a positive integer whose prime factors are all greater than or equal to 7.

Model Proof

1. Preliminaries

For a positive integer N with at least three proper divisors, list all its divisors in increasing order:

$$1 = d_1 < d_2 < d_3 < d_4 < \cdots < d_m = N.$$

Because of the pairing $d_i \cdot d_{m+1-i} = N$, the three largest proper divisors are

$$\frac{N}{d_2}, \frac{N}{d_3}, \frac{N}{d_4}.$$

Hence we can write

$$f(N) \;:=\; \text{sum of the three largest proper divisors of } N \;=\; N\left(\frac{1}{d_2} + \frac{1}{d_3} + \frac{1}{d_4}\right). \qquad (1)$$

C1

C1 |
In the correct formulation, the d_2, d_3, d_4 should be function of N.

The sequence is defined by $a_{n+1} = f(a_n)$, and we require that every term has at least three proper divisors.

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2. Two key lemmas

Lemma 1 (odd numbers).
If N is odd and has at least three proper divisors, then $f(N) < N$ and $f(N)$ is odd.
Proof.
Because all divisors of an odd number are odd, we have $d_2 \geq 3$, $d_3 \geq 5$, $d_4 \geq 7$. Hence

$$\frac{1}{d_2} + \frac{1}{d_3} + \frac{1}{d_4} \;\leq\; \frac{1}{3} + \frac{1}{5} + \frac{1}{7} \;=\; \frac{71}{105} \;<\; 1,$$

so $f(N) < N$. Moreover, each of $N/d_2,\ N/d_3,\ N/d_4$ is odd (odd divided by odd), so their sum is odd. □

Lemma 2 (even numbers not divisible by 3).
If N is even, $3 \nmid N$, and has at least three proper divisors, then
(i) $f(N) < N$, and
(ii) $f(N)$ is **not** a multiple of 6.
Proof.
Since N is even, $d_2 = 2$. The smallest possible values for d_3, d_4 that maximize the sum in (1) are $d_3 = 4$ and $d_4 = 5$ (e.g., $N = 20$). In any case $d_3 \geq 4$ and $d_4 \geq 5$ (because 4 is the only

even number between 2 and 5, and 5 is the smallest integer >4 that does not force a factor 3). Consequently

$$\frac{1}{2}+\frac{1}{d_3}+\frac{1}{d_4} \;\le\; \frac{1}{2}+\frac{1}{4}+\frac{1}{5} \;=\; \frac{19}{20} \;<\; 1,$$

so $f(N) < N$.

We now show that $f(N)$ cannot be divisible by 6.

Write $N = 2^a \cdot m$ with m odd and $3 \nmid m$ ($a \ge 1$). Consider two cases.

Case $a \ge 2$.

Then $4 \mid N$, so $d_3 = 4$. Hence

$$f(N) = \frac{N}{2}+\frac{N}{4}+\frac{N}{d_4} = \frac{3N}{4}+\frac{N}{d_4}.$$

Because $N/4$ is an integer, $3N/4$ is a multiple of 3. Since N is not divisible by 3, N/d_4 is also not divisible by 3. Thus $f(N) \equiv N/d_4 \not\equiv 0 \pmod 3$, and therefore $f(N)$ is not a multiple of 3; in particular it cannot be a multiple of 6.

Case $a = 1$.

Then $N = 2m$ with m odd, $3 \nmid m$. Let p be the smallest prime divisor of m; then $p \ge 5$ and $d_3 = p$. The fourth divisor d_4 is either

- an odd divisor q with $p < q < 2p$ (if such a divisor exists), or
- $2p$ (if no odd divisor lies between p and $2p$).

We examine the two possibilities.

- **Subcase d_4 odd.**
  Then $N/d_4 = 2(m/q)$ is even, $N/2 = m$ is odd, and $N/p = 2(m/p)$ is even. Hence $f(N) = \text{odd} + \text{even} + \text{even} = \text{odd}$, so $f(N)$ is not even and certainly not a multiple of 6.
- **Subcase $d_4 = 2p$ (even).**
  Here

  $$f(N) = \frac{N}{2}+\frac{N}{p}+\frac{N}{2p} = m + 2\frac{m}{p}+\frac{m}{p} = m + 3\frac{m}{p}.$$

  Since m is not divisible by 3, we have $f(N) \equiv m \pmod 3 \neq 0$. Thus $f(N)$ is not a multiple of 3 and therefore not a multiple of 6.

In every subcase $f(N)$ is not a multiple of 6, completing the proof. □

---

3. All terms must be multiples of 6

Assume, for contradiction, that some term a_n is **not** a multiple of 6.

If a_n is odd, Lemma 1 gives that $a_{n+1} = f(a_n)$ is odd and $a_{n+1} < a_n$. Repeating, we obtain an infinite strictly decreasing sequence of positive odd integers - impossible by the well-ordering principle.

If a_n is even but $3 \nmid a_n$, Lemma 2 tells us that $a_{n+1} = f(a_n) < a_n$ and a_{n+1} is **not** a multiple of 6. There are two possibilities for a_{n+1}:

- **a_{n+1} is odd.** Then by Lemma 1 all subsequent terms are odd and strictly decreasing - again impossible.
- **a_{n+1} is even and $3 \nmid a_{n+1}$.** Then we can apply the same reasoning to a_{n+1}: we obtain $a_{n+2} < a_{n+1}$ and a_{n+2} not a multiple of 6, and so on. This would produce an infinite strictly decreasing sequence $a_n > a_{n+1} > a_{n+2} > \dots$ of positive integers, which cannot exist.

Thus, in every case, the assumption leads to a contradiction. Consequently, **every term of the infinite sequence is a multiple of 6.**

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4. Reduction to the form $6M$

Write $a_n = 6M_n$ with $M_n \in \mathbb{N}$. Because $6M_n$ always has at least the divisors $1, 2, 3, 6$, we have $\tau(6M_n) \ge 4$ (i.e., at least three proper divisors).

For $N = 6M$, the three smallest divisors greater than 1 are always $d_2 = 2$ and $d_3 = 3$. The fourth divisor d_4 depends on M:

- **If M is even** $\to 4 \mid 6M$, and since 4 is the smallest integer larger than 3, we have $d_4 = 4$.
- **If M is odd and** $5 \mid M \to 5 \mid 6M$ and $5 < 6$, so $d_4 = 5$.
- **If M is odd and** $5 \nmid M \to$ the next divisor is 6 (because $6 \mid 6M$ and neither 4 nor 5 divides $6M$), so $d_4 = 6$.

Using (1) we compute

$$f(6M) \;=\; \frac{6M}{2} + \frac{6M}{3} + \frac{6M}{d_4} \;=\; 3M + 2M + \frac{6M}{d_4} \;=\; 5M + \frac{6M}{d_4}. \tag{2}$$

Now analyse the three cases.

1. M **odd,** $5 \nmid M \to d_4 = 6$

$$f(6M) = 5M + M = 6M.$$

   Hence $6M$ is a fixed point.
2. M **odd,** $5 \mid M \to d_4 = 5$

$$f(6M) = 5M + \frac{6M}{5} = \frac{31}{5}M.$$

   Since M is odd and divisible by 5, $M/5$ is odd, so the result is odd. Therefore it is **not** a multiple of 6. By the result of §3, such a number cannot appear in an infinite sequence.
3. M **even** $\to d_4 = 4$

$$f(6M) = 5M + \frac{6M}{4} = \frac{13}{2}M.$$

   For this to be a multiple of 6 (necessary for the next term to be admissible) we need

$$\frac{13}{2}M \equiv 0 \pmod 6 \iff 13M \equiv 0 \pmod{12} \iff M \equiv 0 \pmod{12},$$

   C2

   because $13 \equiv 1 \pmod{12}$. Hence, if M is even but **not** divisible by 12, then $f(6M)$ would not be a multiple of 6, contradicting §3. Therefore, in an infinite sequence, every even M must be divisible by 12, and then

$$a_{n+1} = f(6M) = 6 \cdot \left(13 \cdot \frac{M}{12}\right),$$

   so the new parameter is $M' = 13 \cdot (M/12)$.

C2 |

Notation inconsistency. Just use $\nmid$ and $\mid$.

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5. Characterising admissible $M_1 = a_1/6$

Let $M_1 = a_1/6$. We have shown that all terms must be multiples of 6, and the recurrence for the corresponding M_n is:

- If M is odd and $5 \nmid M \to$ fixed point.
- If M is even and divisible by $12 \to M \mapsto 13 \cdot (M/12)$.
- Any other situation leads to a contradiction.

For the sequence to be infinite, we must start from M_1 and, after finitely many applications of the even step[C3], reach an odd M that is not divisible by 5. This forces M_1 to have a very specific structure.

C3 |

even step? 1. Second cases $M \to 13M/12$. 2. even M

Let k be the largest integer such that $12^k \mid M_1$ (the exponent of 12 in M_1). Write

$$M_1 = 12^k \cdot d,$$

where d is a positive integer **not** divisible by 12 (i.e., d is the "remainder" after removing all factors of 12).
We claim that the sequence is infinite **iff** d is **odd** and $5 \nmid d$.

Sufficiency

Assume $M_1 = 12^k \cdot d$ with $k \geq 0$, d odd, and $5 \nmid d$. We prove by induction that

$$M_{i+1} = 13^i \cdot \frac{M_1}{12^i} = 12^{k-i} \cdot 13^i \cdot d \qquad (0 \leq i \leq k).$$

For $i = 0$ this is the definition of M_1.
Inductive step: For $i < k$, we have $M_i = 12^{k-i} \cdot 13^i \cdot d$. Since $k - i \geq 1$, M_i is even and divisible by 12 (because it contains the factor 12^{k-i}). Hence we may apply the even-step rule, obtaining

$$M_{i+1} = 13 \cdot \frac{M_i}{12} = 13 \cdot 12^{k-i-1} \cdot 13^i \cdot d = 12^{k-i-1} \cdot 13^{i+1} \cdot d,$$

which matches the formula.
Thus for $i = k - 1$ we get $M_k = 12^1 \cdot 13^{k-1} \cdot d$, which is even and divisible by 12. Applying the rule once more gives

$$M_{k+1} = 13 \cdot \frac{M_k}{12} = 13^k \cdot d,$$

which is **odd** (since 13^k is odd and d is odd) and not divisible by 5. By case 1 of §4, $6M_{k+1}$ is a fixed point. Consequently,

$$a_{k+1} = 6 \cdot 13^k \cdot d,$$

k+2and the sequence becomes constant thereafter. All terms have at least three proper divisors, so the sequence is infinite.

Necessity

Suppose, for contradiction, that M_1 does **not** have the form $12^k \cdot d$ with d odd and $5 \nmid d$. Write $M_1 = 12^k \cdot d$ as above (now d may be even, or odd but divisible by 5).[C4] We consider two cases.

- d **is even.**[C5]
  Then M_1 is even. If $k = 0$, then M_1 is even but not divisible by 12 (because d even and not a multiple of 12). By §4, case 3, this would give a_2 not a multiple of 6 - contradiction.
  If $k \geq 1$, then M_1 is even and divisible by 12. Apply the even step once to obtain

  $$M_2 = 13 \cdot \frac{M_1}{12} = 13 \cdot 12^{k-1} \cdot d.$$

  Since d is even, M_2 is even. We now check whether M_2 is divisible by 12.
  - If $k - 1 \geq 1$ (i.e., $k \geq 2$), then 12^{k-1} already contains a factor 12, so M_2 is divisible by 12. But then we can continue the process. However, after **exactly** k steps we will obtain

    $$M_{k+1} = 13^k \cdot d,$$

    which is even (because d is even). Since d is not divisible by 12 (by the definition of k), M_{k+1} is **not** divisible by 12. Hence at that moment we would have an even M that is not divisible by 12, which forces $f(6M_{k+1})$ not to be a multiple of 6 - contradiction.
  - If $k = 1$, then $M_1 = 12 \cdot d$ with d even. Then $M_2 = 13d$ is even but not divisible by 12 (because $13d$ lacks the necessary factor 3 and[C6] the factor 4 if d is not a multiple of 12). Again, M_2 fails the requirement that every even M must be divisible by 12.

  In all subcases, contradiction arises.
- d **is odd but** $5 \mid d$**.**
  Then after k applications of the even step we obtain

  $$M_{k+1} = 13^k \cdot d,$$

C4 |
This formulation should mention $12 \nmid d$

C5 |
No need to consider for even or odd, we can directly say $M_{k+1} = 13^k \cdot d$ and then contradictory. We already have the dynamic of M_i at the beginning.

C6 |
or

which is odd and divisible by 5. Consequently, $a_{k+1} = 6M_{k+1}$ falls into case 2 of §4, yielding $f(a_{k+1})$ odd. As in §3, an odd term would lead to an infinite strictly decreasing sequence of odd numbers - contradiction.

Thus, the only way to avoid contradictions is that d is odd and $5 \nmid d$.

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6. Final answer

Combining everything, we obtain that the initial value a_1 can be exactly those numbers of the form

$$\boxed{a_1 = 6 \cdot 12^k \cdot d \qquad (k \in \mathbb{N}_0,\ d \text{ odd},\ 5 \nmid d).}$$

Equivalently, $a_1 = 2^{2k+1} \cdot 3^{k+1} \cdot d$ with d odd and not divisible by 5.

Human Expert Comment

$\boxed{7/7}$

Problem 5

Alice and Bazza are playing the inekoalaty game, a two-player game whose rules depend on a positive real number λ which is known to both players. On the nth turn of the game (starting with $n = 1$) the following happens:

- If n is odd, Alice chooses a nonnegative real number x_n such that

$$x_1 + x_2 + \cdots + x_n \leq \lambda n.$$

- If n is even, Bazza chooses a nonnegative real number x_n such that

$$x_1^2 + x_2^2 + \cdots + x_n^2 \leq n.$$

If a player cannot choose a suitable x_n, the game ends and the other player wins. If the game goes on forever, neither player wins. All chosen numbers are known to both players.
Determine all values of λ for which Alice has a winning strategy and all those for which Bazza has a winning strategy.

Short Answer
The outcome of the inekoalty game depends on the value of λ as follows:

- If $\lambda > \frac{\sqrt{2}}{2}$, Alice has a winning strategy.
- If $\lambda < \frac{\sqrt{2}}{2}$, Bazza has a winning strategy.
- If $\lambda = \frac{\sqrt{2}}{2}$, neither player has a winning strategy, and the game can go on forever.

Model Proof

Notation and basic facts
Define $S_n = \sum_{i=1}^{n} x_i$ and $Q_n = \sum_{i=1}^{n} x_i^2$.
The rules: on odd n, $S_n \leq \lambda n$; on even n, $Q_n \leq n$.
From Cauchy-Schwarz, after any even turn $2k$ we have

$$S_{2k}^2 \leq (2k)Q_{2k} \leq (2k) \cdot 2k = (2k)^2 \quad \Longrightarrow \quad S_{2k} \leq 2k. \tag{1}$$

For $k \geq 1$ define the **slack** after the even turn $2k$:

$$d_k = \lambda(2k+1) - S_{2k}.$$

Alice can move on turn $2k+1$ iff $d_k \geq 0$ (she may choose $x_{2k+1} = 0$).
If at some odd turn Alice can legally choose a number $u > \sqrt{2}$ while $Q_{2k} = 2k$, then $Q_{2k+1} = 2k + u^2 > 2k + 2$ and Bazza will have no legal move on the next turn, so Alice wins immediately.
In the general case, we will use the slack to analyse the game.

Key strategy for Bazza
We consider the following natural strategy for Bazza on even turns: after Alice's move u on turn $2k+1$, if the game has not ended, Bazza chooses

$$v = \sqrt{2k + 2 - Q_{2k+1}},$$

i.e. the largest possible number that still satisfies $Q_{2k+2} \leq 2k + 2$.
This choice is always legal as long as $Q_{2k+1} \leq 2k + 2$ and makes $Q_{2k+2} = 2k + 2$ (the

maximal possible sum of squares).
We call this the **maximal strategy** for Bazza.

Claim 1. As long as Alice never picks a number $u > \sqrt{2}$ (which would make her win immediately), the maximal strategy forces $Q_{2k} = 2k$ for every $k \geq 1$.

Proof. By induction. For $k = 1$, after Alice's first move $x_1 = a$ (with $0 \leq a \leq \lambda$), Bazza can choose $v = \sqrt{2 - a^2}$ because $a^2 \leq 2$ (this holds for all relevant λ; in the cases where we apply it, we have $\lambda \leq \sqrt{2}/2 < \sqrt{2}$, and for larger λ Alice would win earlier). Then $Q_2 = a^2 + v^2 = 2$. Inductive step: assume $Q_{2k} = 2k$. Alice's move u satisfies $Q_{2k+1} = 2k + u^2 \leq 2k + 2$ because otherwise $u^2 > 2$ and Alice would have won (this can only happen if $Q_{2k} = 2k$ and $u > \sqrt{2}$).[D1] If the game continues, we must have $u \leq \sqrt{2}$. Then $Q_{2k+1} = 2k + u^2 \leq 2k + 2$. Bazza chooses $v = \sqrt{2k + 2 - (2k + u^2)} = \sqrt{2 - u^2}$, yielding $Q_{2k+2} = 2k + 2$. □

Do not understand. Reads like a thinking draft process.

Analysis of the slack under the maximal strategy

Suppose the maximal strategy is followed and that the game has not ended (so far Alice has never picked $u > \sqrt{2}$). Let $u = x_{2k+1}$ and $v = x_{2k+2}$. Then

$$S_{2k+2} = S_{2k} + u + v, \qquad Q_{2k+2} = 2k + 2,$$

and therefore

$$d_{k+1} = \lambda(2k + 3) - S_{2k+2} = d_k + 2\lambda - (u + v). \tag{2}$$

For $0 \leq u \leq \sqrt{2}$ define $h(u) = u + \sqrt{2 - u^2}$. One checks that $h(u) \geq \sqrt{2}$, with equality exactly at $u = 0$ and $u = \sqrt{2}$, and maximum 2 at $u = 1$. Consequently

$$d_{k+1} \leq d_k + 2\lambda - \sqrt{2}, \tag{3}$$

and equality can be achieved by choosing $u = 0$ (or $u = \sqrt{2}$, if allowed).
Thus the maximal possible increase of d_k (when Alice tries to keep her slack large) is $2\lambda - \sqrt{2}$, attained by taking $u = 0$.

Three regimes

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1. $\lambda < \dfrac{\sqrt{2}}{2}$

Then $2\lambda - \sqrt{2} < 0$.

Bazza's winning strategy.
Bazza adopts the maximal strategy described above.
First, compute an upper bound for d_1. After the first two moves, $S_2 = a + \sqrt{2 - a^2}$ with $0 \leq a \leq \lambda$[D2]. The function $f(a) = a + \sqrt{2 - a^2}$ satisfies $f(a) \geq \sqrt{2}$ (minimum at $a = 0$), so

D2 |
Better to use x_1 not a

$$d_1 = 3\lambda - S_2 \leq 3\lambda - \sqrt{2}. \tag{4}$$

Because $\lambda < \sqrt{2}/2$, we have $3\lambda - \sqrt{2} < \sqrt{2}/2 < \sqrt{2}$.
Now, from (3) we obtain for any possible play

$$d_k \leq d_1 + (k - 1)(2\lambda - \sqrt{2}) \leq (3\lambda - \sqrt{2}) + (k - 1)(2\lambda - \sqrt{2}). \tag{5}$$

(The first inequality follows by induction using $d_{k+1} \leq d_k + 2\lambda - \sqrt{2}$, the second uses the bound on d_1.)
Since $2\lambda - \sqrt{2} < 0$, the right-hand side of (5) tends to $-\infty$ as $k \to \infty$. Hence there exists a finite K with $d_K < 0$.

When $d_K < 0$, at the beginning of turn $2K + 1$ Alice cannot move (even $x = 0$ would violate the sum condition), so Bazza wins.

Moreover, we claim that Alice never gets the chance to win earlier by picking $u > \sqrt{2}$. Indeed, from (5) and $2\lambda - \sqrt{2} < 0$, we have $d_k \leq 3\lambda - \sqrt{2} < \sqrt{2}$ for all k. Hence d_k never exceeds $\sqrt{2}$, so she cannot legally pick a number larger than $\sqrt{2}$ (which would be required to win immediately). Thus Bazza's strategy is winning.

Conclusion for $\lambda < \sqrt{2}/2$: Bazza has a winning strategy.

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2. $\lambda = \dfrac{\sqrt{2}}{2}$

Here $2\lambda - \sqrt{2} = 0$.

Bazza does not have a winning strategy.

We exhibit a strategy for Alice that prevents Bazza from ever winning.

Let Alice always choose $x_1 = 0$ and thereafter $x_{2k+1} = 0$ on every odd turn.

Consider any Bazza moves. On even turns we have only Bazza's numbers, say $y_1, \ldots, y_k$ at turn $2k$. The constraints are $\sum_{i=1}^{k} y_i^2 = Q_{2k} \leq 2k$ (because odd terms are zero). By Cauchy-Schwarz,

$$S_{2k} = \sum_{i=1}^{k} y_i \leq \sqrt{k \sum_{i=1}^{k} y_i^2} \leq \sqrt{k \cdot 2k} = \sqrt{2}\, k.$$

Therefore

$$d_k = \lambda(2k+1) - S_{2k} \geq \frac{\sqrt{2}}{2}(2k+1) - \sqrt{2}\, k = \frac{\sqrt{2}}{2} > 0.$$

Thus Alice can always move (she may choose 0). Also, she never wins because to win she would need to make $Q_{2k+1} > 2k+2$, which would require $x_{2k+1}^2 > 2k+2-Q_{2k} \geq 2$, i.e. $x_{2k+1} > \sqrt{2}$. But $x_{2k+1} \leq d_k$ and we have not shown an upper bound on d_k;[D4] however, the crucial point is that with this Alice strategy, Bazza never wins because Alice never loses. So this suffices to prove that Bazza does not have a winning strategy (he cannot force a win against every Alice strategy).

D4 |

This conclusion only focuses on a specific B's policy, which already discussed at the beginning. This looks like thinking draft process.

Alice does not have a winning strategy.

We now show that Bazza has a strategy to prevent Alice from winning.

Let Bazza adopt the maximal strategy (as in Case 1). We prove that under this strategy, Alice can never win.

First, by Claim 1, as long as the game continues, $Q_{2k} = 2k$. Using (3) with $\lambda = \sqrt{2}/2$ we get

$$d_{k+1} = d_k + \sqrt{2} - (u+v) \leq d_k, \tag{6}$$

because $u + v \geq \sqrt{2}$. Hence d_k is non-increasing. Moreover, from (4) we have

$$d_1 \leq 3\lambda - \sqrt{2} = \sqrt{2}/2.$$

Thus $d_k \leq \sqrt{2}/2 < \sqrt{2}$ for all k. In particular, d_k never exceeds $\sqrt{2}$, so Alice can never pick a number $u > \sqrt{2}$ (which would be necessary to win immediately).

Could Alice win by picking some $u \leq \sqrt{2}$ and later build up? To win, she must at some odd turn make $Q_{2k+1} > 2k+2$. Since $Q_{2k} = 2k$, this would require $u^2 > 2$, i.e. $u > \sqrt{2}$. Thus any winning move would necessarily be $u > \sqrt{2}$. Because $d_k \leq \sqrt{2}/2 < \sqrt{2}$, such a move is impossible. Therefore Alice never wins.[D5]

D5 |

??? Reads like a thinking draft process.

Additionally, we check that d_k never becomes negative, so Alice always can move. Indeed, using (2) and the fact that $u \leq d_k$ (so $u \leq \sqrt{2}/2 < 1$), the smallest possible d_{k+1} for a given d_k is attained when $u = d_k$, giving $d_{k+1} = \sqrt{2} - \sqrt{2 - d_k^2} \geq 0$. Hence $d_k \geq 0$ for all k.[D6]

D6 |

No need. The proof is finished.

Thus the maximal strategy by Bazza prevents Alice from winning, and also avoids his own loss (the game can continue indefinitely, e.g., if Alice always chooses 0, then $d_k = \sqrt{2}/2$ constant and $Q_{2k} = 2k$, so the play never ends). Consequently, Alice cannot force a win; she has no winning strategy.

Since neither player has a strategy that forces a win, the game is a draw for $\lambda = \sqrt{2}/2$.

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3. $\lambda > \dfrac{\sqrt{2}}{2}$

Now $2\lambda - \sqrt{2} > 0$.

Alice's winning strategy.

She will always choose 0 on her odd turns, until she can force a win.

Specifically, on turn 1 she sets $x_1 = 0$. On each subsequent odd turn $2k+1$ ($k \geq 1$), she examines the current slack $d_k = \lambda(2k+1) - S_{2k}$. If $d_k > \sqrt{2k+2-Q_{2k}}$, she chooses a

number u with

$$\sqrt{2k+2-Q_{2k}} < u \le d_k,$$

such that $u^2 > 2k+2-Q_{2k}$ (this is possible because d_k is strictly larger). She then wins on the next turn because $Q_{2k+1} = Q_{2k} + u^2 > 2k+2$. If $d_k \le \sqrt{2k+2-Q_{2k}}$, she simply chooses $x_{2k+1} = 0$.

We must verify that this strategy is legal and that it indeed leads to a win.

First, note that as long as Alice always picks 0, all her moves contribute nothing to Q and S_{2k} is the sum of Bazza's even moves. For any Bazza strategy we have

$$Q_{2k} = \sum_{i=1}^{k} x_{2i}^2 \le 2k, \qquad S_{2k} = \sum_{i=1}^{k} x_{2i}.$$

By Cauchy-Schwarz,

$$S_{2k}^2 \le k\,Q_{2k} \le 2k^2 \quad \Longrightarrow \quad S_{2k} \le \sqrt{2}\,k. \tag{7}$$

Consequently,

$$d_k = \lambda(2k+1) - S_{2k} \ge \lambda(2k+1) - \sqrt{2}\,k = (2\lambda - \sqrt{2})k + \lambda. \tag{8}$$

Because $2\lambda - \sqrt{2} > 0$, the right-hand side of (8) tends to $+\infty$ as $k \to \infty$. In particular, there exists an integer K such that

$$d_K > \sqrt{2K+2-Q_{2K}}.$$

(Since the right side is at most $\sqrt{2K+2}$, it suffices that $(2\lambda - \sqrt{2})K + \lambda > \sqrt{2K+2}$; this holds for all sufficiently large K.)

At the start of turn $2K+1$, we have $d_K > \sqrt{2K+2-Q_{2K}}$. Choose

$$u = \sqrt{2K+2-Q_{2K}} + \varepsilon,$$

with $\varepsilon > 0$ small enough that $u \le d_K$ (possible because d_K is strictly larger). Then

$$S_{2K+1} = S_{2K} + u \le S_{2K} + d_K = \lambda(2K+1),$$

so the sum condition is satisfied, and

$$Q_{2K+1} = Q_{2K} + u^2 > Q_{2K} + (2K+2-Q_{2K}) = 2K+2.$$

Therefore on turn $2K+2$ Bazza has no legal move (even $x = 0$ would give $Q > 2K+2$). Alice wins.

It remains to check that Alice can always make the choice $x_{2k+1} = 0$ when $d_k \le \sqrt{2k+2-Q_{2k}}$. The condition for choosing 0 is that $0 \le d_k$, i.e. $d_k \ge 0$. From (8), $d_k \ge (2\lambda - \sqrt{2})k + \lambda > 0$ for all k (since the right side is positive for $k = 1$ because $3\lambda - \sqrt{2} > 0$ for $\lambda > \sqrt{2}/2$). Hence Alice can always pick 0 when she does not want to win yet. Thus her strategy is well defined and guarantees a win in finite time.

Conclusion for $\lambda > \sqrt{2}/2$: Alice has a winning strategy.

Final answer

- Alice has a winning strategy **iff** $\lambda > \dfrac{\sqrt{2}}{2}$.
- Bazza has a winning strategy **iff** $\lambda < \dfrac{\sqrt{2}}{2}$.

- For $\lambda = \frac{\sqrt{2}}{2}$ neither player can force a win; the game can be drawn.

Human Expert Comment

Some parts of the proof read like a thinking process. It feels like it realizes that c has not been proved, then decides that proving c is not actually necessary. The proof seems finished, but then it starts thinking again.

$\boxed{7/7}$